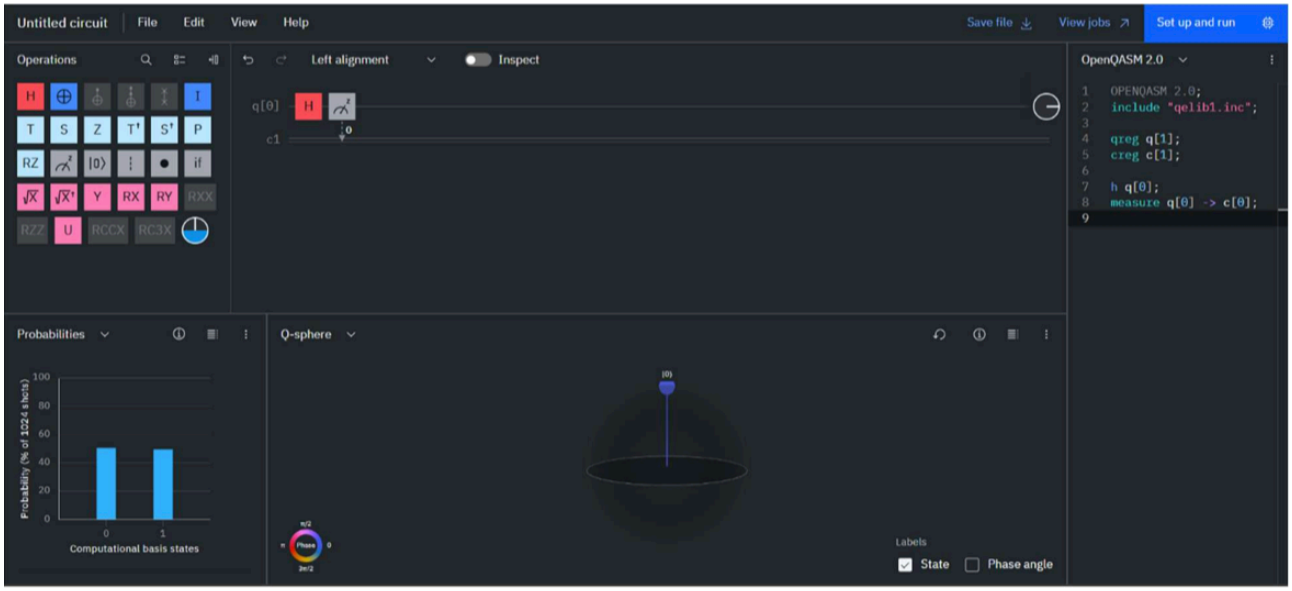
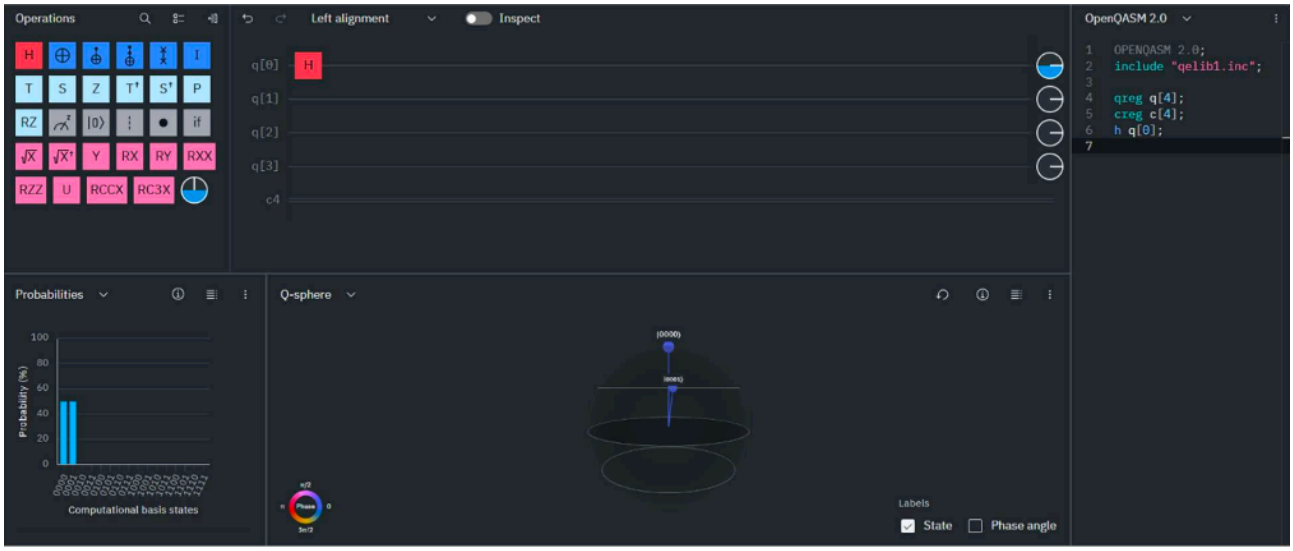
QC Assignment - 2

Q2.

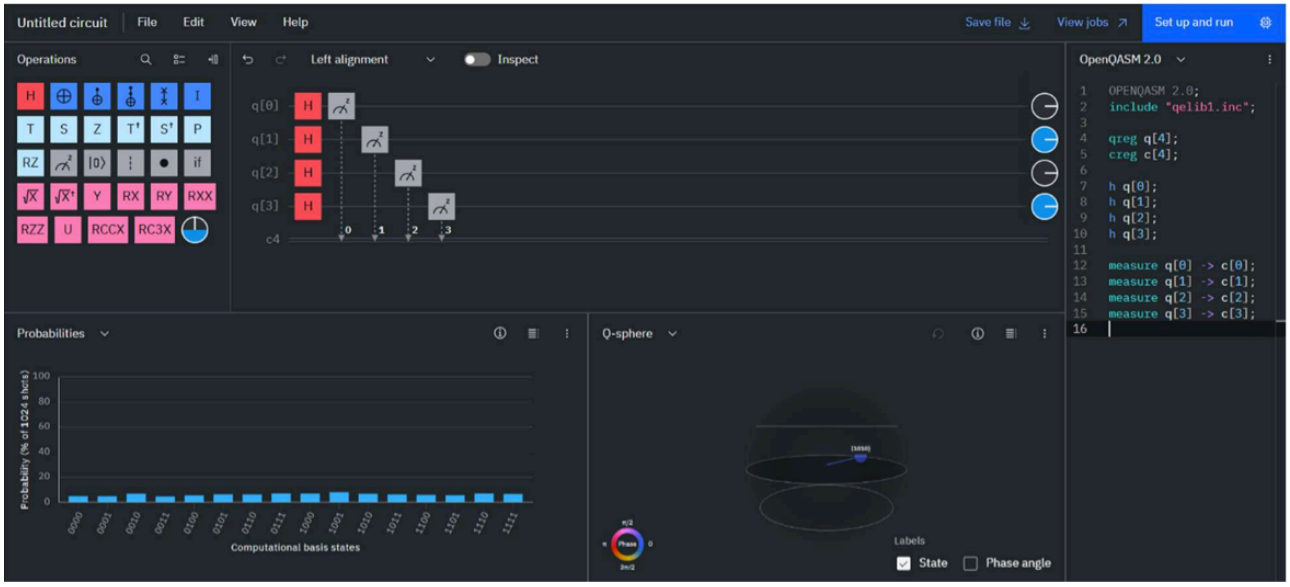


Q3.

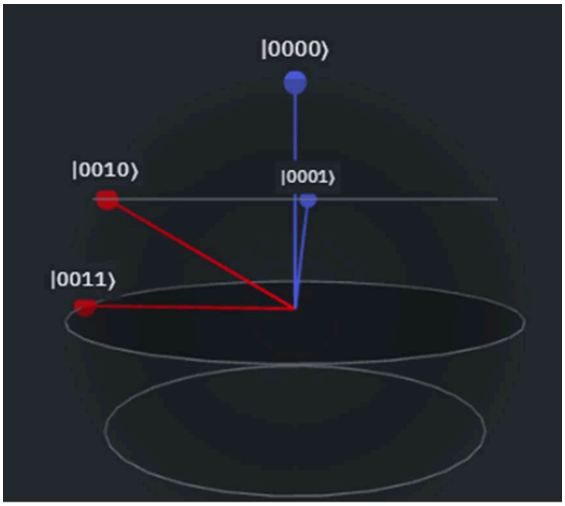


When you run a circuit with a single Hadamard gate, the qubit enters a **superposition state**. Every time you measure it (single shot), you will get either a 0 or a 1 with equal probability (50% each). It is perfectly random.

Q4.



Q5.



Q6.

```
import cmath
z = 1 + 1j

r,theta = cmath.polar(z)
print("Polar form (r,θ) : ",r,theta)

cartesian = cmath.rect(r,theta)
print("Cartesian form : ",cartesian)
```

```
Polar form (r,θ) :  1.4142135623730951 0.7853981633974483
Cartesian form :  (1.0000000000000002+1j)
```